To find T_3 put r = 2 in T_{r+1}

Here $n = 5,\ x = 3x,\ a = y$

$$T_{r+1} = nC_r\, x^{n-r}\, a^r$$

$$T_{2+1} = 5C_2\, (3x)^{5-2}\, (y)^2$$

$$= 5C_2\, 3^3\ x^3\, y^2$$

$$= 270x^3y^2$$

Similarly we can find by putting $r = 3$ in T_{r+1} to get T_4, then $T_4 = 90x^2y^3$

Binomial theorem for a rational index:

For any rational number other than positive integer

$$(1+x)^n = 1 + \frac{n}{1!}x + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \ldots, \quad \text{provided } |x| < 1.$$

NOTE

(i) If $n \in N$, $(1+x)^n$ is desired for all value of x and if n is a rational number other than the natural number then $(1+x)^n$ is desired only when $|x| < 1$.

(ii) If $n \in N$, then the expansion of $(1+x)^n$ contains only $(n+1)$ terms. If n is a rational number, then the expansion of $(1+x)^n$ contain infinitely many terms.

Some important expansions:

$$(1-x)^n = 1 - \frac{n}{1!}x + \frac{n(n-1)}{2!}x^2 - \frac{n(n-1)(n-2)}{3!}x^3 + \ldots$$

$$(1+x)^{-n} = 1 - \frac{n}{1!}x + \frac{n(n+1)}{2!}x^2 - \frac{n(n+1)(n+2)}{3!}x^3 + \ldots$$

$$(1-x)^{-n} = 1 + \frac{n}{1!}x + \frac{n(n+1)}{2!}x^2 + \frac{n(n+1)(n+2)}{3!}x^3 + \ldots$$

Special cases of infinite series:

$$(1+x)^{-1} = 1 - x + x^2 - x^3 + \ldots$$

$$(1-x)^{-1} = 1 + x + x^2 + x^3 + \ldots$$

$$(1+x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + \ldots$$

$$(1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \ldots$$